

Em 16
 Con Macan hte 6 HW

Review

1 $^{15}C_{11} = 1365$

2 $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} = \tan \frac{\pi}{6} \quad (c)$

3 $(x-2)^2 + (y+3)^2 = r^2$

$y = 2x + 3$

$y = -\frac{1}{2}x + c$ perpendicular (3, -3)

$y + 3 = -\frac{1}{2}(x - 2)$

$y = -\frac{1}{2}x - 2$

$y = 2x + 3$ intersects with $y = -\frac{1}{2}x - 2$:

$2x + 3 = -\frac{1}{2}x - 2$

$\frac{5}{2}x = -5$

$x = -2$

$y = -1$

distance from (3, -3) to (-2, -1):

$= \sqrt{4^2 + 2^2}$

$= \sqrt{20} \quad (a)$

4 $\frac{1}{x-2} + 1$

$= \frac{1 - \frac{1}{x+2}}{x+2} \quad (b)$

5 $y = \tan 2x$

$y' = 2 \sec^2 2x$

At $x=0$: $y' = 2 \sec^2 0 = \frac{2}{\cos^2 0} = \underline{2}$

Consolidation

$$\begin{aligned} \int \frac{1+2x}{\sqrt{x+2x^2}} dx &= \int \frac{1+2x}{\sqrt{u}} \times \frac{1}{1+2x} du \\ &= \int u^{-\frac{1}{2}} du \end{aligned}$$

$= 2u^{\frac{1}{2}}$

$= 2\sqrt{x+2x^2}$

$$2) \int x(3-2x)^9 dx$$

$$= \int \frac{v-3}{2} \times v^9 \times \frac{1}{2} dv \quad v = 3-2x$$

$$= \int \frac{v^2(v-3)}{4} dv$$

$$= \frac{1}{4} \int v^3(v-3) dv$$

$$= \frac{1}{4} \int v^4 - 3v^3 dv$$

$$= \frac{1}{20} v^5 - \frac{3}{8} v^4 + C$$

$$= \frac{(3-2x)^5}{20} - \frac{(3-2x)^4}{8} + C$$

$$6) \int \frac{x}{\sqrt{1+x}} dx$$

$$= \int \frac{v-1}{v^{\frac{1}{2}}} \times 1 dv$$

$$= \int v^{\frac{1}{2}} - v^{-\frac{1}{2}} dv$$

$$= \frac{2}{3} v^{\frac{3}{2}} + 2 v^{\frac{1}{2}} + C$$

$$= \frac{2}{3} (1+x)^{\frac{3}{2}} + 2 (1+x)^{\frac{1}{2}} + C$$

$$3) a) \int_2^9 (v-2)v^{\frac{1}{2}} dv \quad v = 2+x$$

$$= \int_4^9 v^{\frac{3}{2}} - 2v^{\frac{1}{2}} dv$$

$$= \left[\frac{2}{5} v^{\frac{5}{2}} - \frac{4}{3} v^{\frac{3}{2}} \right]_4^9$$

$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{4}{3} \times 9^{\frac{3}{2}} - \frac{2}{5} \times 4^{\frac{5}{2}} + \frac{4}{3} \times 4^{\frac{3}{2}}$$

$$= \frac{2}{5} \times 3^5 - \frac{4}{3} \times 3^3 - \frac{2}{5} \times 2^5 + \frac{4}{3} \times 2^3$$

$$= \frac{886}{15}$$

$$6) \int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{1+\cos \theta} d\theta$$

$$v = 1 + \cos \theta$$

$$= \int_2^1 \frac{2 \sin \theta \cos \theta}{v} \times \frac{1}{-\sin \theta} dv$$

$$= \int_2^1 - \frac{2 \cos \theta}{v} dv$$

$$= \int_2^1 \frac{2(1-v)}{v} dv$$

$$= \int_2^1 2v^{-1} - 2 dv$$

$$= [2 \ln |v| - 2v]_2^1$$

$$= -2 - 2 \ln 2 + 4$$

$$= 2 - 2 \ln 2$$

$$4) \int \sec^2 x \tan x \sqrt{1+\tan x} dx$$

$$v^2 = 1 + \tan x$$

$$\int \sec^2 x \tan x v \times \frac{2}{2 \sec^2 x} dv$$

$$\int 2(v^2-1)v dv$$

$$\int 2v^3 - 2v dv$$

$$= \frac{1}{2} v^4 - v^2 + C$$

$$= \frac{1}{2} (1+\tan x)^2 - 1 + \tan x + C$$

$$b) \int \sec^4 x dx$$

$$= \int (1+\tan^2 x)(\sec^2 x) dx$$

$$= \int (1+v^2)(\sec^2 x) \times \frac{1}{\sec^2 x} dv$$

$$= \int 1+v^2 dv$$

$$= v + \frac{1}{3} v^3 + C$$

$$= \tan x + \frac{\tan^3 x}{3} + C$$

$$5 \int_0^{\frac{\pi}{3}} \sin^3 x \cos^2 x \, dx$$

$$\int_1^{\frac{1}{2}} u^2(u^2-1) \, du$$

$$= \left[\frac{1}{5} u^5 - \frac{1}{3} u^3 \right]_1^{\frac{1}{2}}$$

$$= \left(\frac{1}{5} - \frac{1}{3} - \frac{1}{160} + \frac{1}{24} \right)$$

$$= \frac{47}{480}$$

$$6 \int_6^{20} \frac{8x}{1+x+1} \, dx$$

$$\int_5^9 \frac{2v^2-2}{v} \times \frac{v}{2} \, dv \quad u=4x+1 \quad \frac{du}{dx}=4$$

$$\frac{dv}{dx} = \frac{2}{v}$$

$$\frac{dz}{dx} = \frac{v}{2}$$

$$\int_5^9 (v^2-1) \, dv$$

$$\left[\frac{1}{3} v^3 - v \right]_5^9$$

$$= \frac{729}{3} - 9 - \frac{125}{3} + 5$$

$$= \frac{604}{3} - 4$$

$$= \frac{592}{3}$$

$$7) \int 2 \sin^2 x \, dx \quad \cos 2x = \cos^2 x - \sin^2 x$$

$$= \int \sin^2 x - \cos^2 x + \sin^2 x + \cos^2 x \, dx$$

$$= \int 1 - \cos 2x \, dx$$

$$= x - \sin 2x + c$$

$$ii) \int 3 \cos^3 x \, dx$$

$$\int 3 \cos x \cos^2 x \, dx$$

$$\int 3 \cos x - 3 \sin^2 x \cos x \, dx$$

$$= 3 \sin x - \sin^3 x + c$$

$$\begin{aligned} \text{iii)} \int 4 \tan x \, dx \\ = \int 4 \frac{\sin x}{\cos x} \, dx \\ = -4 \ln |\cos x| + c \end{aligned}$$

$$\begin{aligned} 8 \text{ i)} \int \cot^2 2x \, dx &= \int \frac{\cos^2 2x}{\sin^2 2x} \, dx \\ &= \int \frac{1 - \sin^2 2x}{\sin^2 2x} \, dx \\ &= \int \csc^2 2x - 1 \, dx \\ &= -\frac{1}{2} \cot 2x - x + c \end{aligned}$$

$$\begin{aligned} \text{ii)} \int \frac{\tan x}{\sin 2x} \, dx \\ = \int \frac{\frac{\sin x}{\cos x}}{2 \sin x \cos x} \, dx \\ = \int \frac{1}{2 \cos^2 x} \, dx \\ = \int \frac{1}{2} \sec^2 x \\ = \frac{1}{2} \tan x + c \end{aligned}$$

$$\begin{aligned} \text{iii)} \int \sin^2 3x \, dx \\ = \int \frac{1 - \cos 6x}{2} \, dx \\ = \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + c \\ = \frac{1}{2} x - \frac{1}{12} \sin 6x + c \end{aligned}$$

$$\begin{aligned} \text{iv)} \int (\sin 2x + \cos 2x)^2 \, dx \\ \int \sin^2 2x + 2 \sin 2x \cos 2x + \cos^2 2x \, dx \\ = \int 1 + \sin 4x \, dx \\ = x - \frac{1}{4} \cos 4x + c \end{aligned}$$

$$\begin{aligned} \text{v)} \int \cos^2 x \sin x \, dx \\ = -\frac{1}{3} \cos^3 x + c \end{aligned}$$

$$\begin{aligned} 10 - \int \frac{1}{\sqrt{1-\cos \theta}} \times \sin \theta \, d\theta \quad \begin{matrix} x = \cos \theta \\ \frac{dx}{d\theta} = -\sin \theta \end{matrix} \\ \int 1 \, d\theta \end{aligned}$$

$$= \theta + C$$

$$= \arccos x + C \quad \left(\begin{array}{l} \cos \theta = x \\ \therefore x = \arccos \theta \end{array} \right)$$

$$11 \int_{\ln 3}^{\ln 4} \frac{e^{4x}}{e^x - 2} dx$$

$$\int_1^{\sqrt{2}} \frac{e^{4x}}{u^2} \times \frac{2}{e^x} du$$

$$\int_1^{\sqrt{2}} \frac{2e^{3x}}{u^2} du$$

$$= \int_1^{\sqrt{2}} \frac{2(u^2 + 2)^3}{u} du$$

$$= \int_1^{\sqrt{2}} 2u^5 + 12u^3 + 24u + 16u^{-1} du$$

$$= \left[\frac{2}{6} u^6 + 3u^4 + 12u^2 + 16 \ln u \right]_1^{\sqrt{2}}$$

$$= \frac{4}{3} + 12 + 24 + 16 \ln \sqrt{2} - \frac{1}{3} - 3 - (2 + 16 \ln 1)$$

$$= \frac{7}{3} + 21 + 16 \ln \sqrt{2}$$

$$= \frac{70}{3} + 16 \ln \sqrt{2}$$

$$= \frac{70}{3} + 8 \ln 2$$

$$\therefore a=70, b=3, c=8, d=2$$

Exam Question

$$1) a) \int 5x^3 \sqrt{5x^2 + 1} dx \quad u = 5x^2 + 1$$

$$\int \frac{5}{2}(u-1)\sqrt{u} du$$

$$\int \frac{5}{2}u^{\frac{3}{2}} - \frac{5}{2}u^{\frac{1}{2}} du$$

$$= u^{\frac{5}{2}} - \frac{5}{3}u^{\frac{3}{2}} + C$$

$$b) \int x \tan^2 x dx$$

$$= \int x (\sec^2 x - 1) dx$$

$$= \int x \sec^2 x dx - \int x dx$$

$$= x \tan x - \int \tan x dx - \frac{1}{2}x^2$$

$$= x \tan x - \ln |\sec x| - \frac{1}{2}x^2 + C$$

$$2 \int e^{\sqrt{x+1}} dx$$

$$\int e^u \cdot 2u du$$

$$= 2(e^u u - \int e^u du)$$

$$= 2(e^u u - e^u)$$

$$= 2e^{\sqrt{x+1}} \sqrt{x+1} - e^{\sqrt{x+1}} + C$$

$$3 \int_{\frac{1}{2}\pi}^{\frac{3}{2}\pi} (\sec 2x + \tan 2x)^2 dx$$

$$\int_{\frac{1}{2}\pi}^{\frac{3}{2}\pi} \sec^2 2x + 2\sec 2x \tan 2x + \tan^2 2x dx$$

$$= \left[\frac{1}{2} \tan 2x + \frac{1}{2} \tan(2x) - x \right]_{\frac{1}{2}\pi}^{\frac{3}{2}\pi} + 2 \int_{\frac{1}{2}\pi}^{\frac{3}{2}\pi} \frac{\sin 2x}{\cos^2 2x} dx$$

$$=$$

$$2 \int \sin 2x \times (\cos 2x)^{-2} dx$$

$$2 \left(\sin 2x \times \frac{1}{2} \tan 2x - \int \frac{\sin 2x}{\cos 2x} \times \cos 2x dx \right)$$

$$2 \left(\frac{1}{2} \cos 2x + \frac{1}{2} \sin 2x \tan 2x \right)$$

$$= \left[-x + \tan 2x + \cos 2x + \sin 2x \tan 2x \right]_{\frac{1}{2}\pi}^{\frac{3}{2}\pi}$$

$$= -\frac{\pi}{2} + 2$$